

Software Requirements Specification

ShopEase LK — Trusted Online Grocery & General Commerce Platform

Document type	Version	Date	Prepared by
SRS	v1.0	May 2026	R.M..T.D.Rathnayake

Attribute	Detail
Project	ShopEase LK E-Commerce Platform
Organization	ShopEase LK (Startup — Sri Lanka)
BA Role	
Supervisor	
Status	Draft — v1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the complete functional and non-functional requirements for the ShopEase LK e-commerce platform. It is intended to serve as the authoritative technical reference for developers, testers, and project stakeholders throughout the development lifecycle. This document translates the business requirements defined in the ShopEase LK BRD (v1.0) into precise, testable system requirements.

1.2 Project Background

ShopEase LK is a startup e-commerce initiative targeting the Sri Lankan general consumer market. A contextual analysis of Sri Lankan consumer behavior identified three root causes preventing online shopping adoption: (1) lack of trust in product quality, (2) high and unreliable delivery costs, and (3) the absence of cash-on-delivery options. ShopEase LK addresses these through seller verification, a hybrid delivery model integrating PickMe/Uber API for urgent orders, a pickup-point network for economy delivery, and cash-on-delivery across all delivery methods.

1.3 Intended Audience

Audience	How they use this document
Development team	Build system features based on functional requirements
QA / Test team	Derive test cases from each FR and NFR
Business Analyst	Maintain traceability between BRD and SRS
CEO / Product Owner	Validate requirements against business objectives
IT Manager	Review architecture and interface specifications

1.4 Scope

This SRS covers the ShopEase LK MVP (Phase 1) — a web-based e-commerce platform for general products and groceries in Sri Lanka. It includes all functional requirements, non-functional requirements, use case diagrams, use case scenarios, activity diagrams, and data design for the MVP scope. Mobile applications, AI recommendations, multi-language support, and subscription baskets are explicitly out of scope for Phase 1.

1.5 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification — this document
BRD	Business Requirements Document — the parent document
FR	Functional Requirement — what the system must do
NFR	Non-Functional Requirement — how well the system must perform
UC	Use Case — a scenario describing actor-system interaction
COD	Cash on Delivery — payment collected at delivery point
MVP	Minimum Viable Product — Phase 1 launch version
API	Application Programming Interface — PickMe/Uber integration
ERD	Entity Relationship Diagram — database design model
LKR	Sri Lankan Rupee — the currency used on the platform
Admin	ShopEase platform administrator
Shall	Mandatory requirement (IEEE standard terminology)
Should	Recommended but not mandatory requirement
May	Optional requirement

2. Overall Description

2.1 Product Perspective

ShopEase LK is a new, standalone web-based platform. It integrates with two external systems: (1) the PickMe/Uber delivery API for real-time courier dispatch and tracking, and (2) a third-party LKR payment gateway for card transactions. The platform has four primary interfaces: customer-facing storefront, seller dashboard, admin panel, and delivery partner API integration layer.

2.2 Product Functions (High-Level Summary)

- Customer registration, login, and profile management
- Product browsing, search, and filtering
- Seller registration, verification, and product listing management
- Shopping cart, wishlist, and checkout
- Hybrid delivery selection: PickMe/Uber (urgent), ShopEase delivery (standard), pickup points (economy)
- Card payment and cash-on-delivery on all delivery methods
- Real-time order tracking and automated notifications
- Product reviews with photos and videos
- Admin panel: seller approval, order management, dispute resolution

2.3 User Classes and Characteristics

User class	Characteristics	Technical ability
Customer	General public buying groceries and products online	Basic internet user
Seller / Vendor	Small-medium retailers listing products for sale	Moderate — can use web forms
Admin	ShopEase staff managing the platform	High — full system access
Delivery Partner	PickMe/Uber API (system actor, not human)	System integration
Pickup Point Agent	Staff at partner shops receiving and handing orders	Low — simple confirmation screen

2.4 Operating Environment

- Platform type: Web application (responsive design for desktop and mobile browsers)
- Hosting: Cloud-based (AWS or Azure) for scalability
- Browser support: Chrome, Firefox, Safari, Edge (latest 2 versions)
- Mobile browser: Chrome and Safari on Android and iOS
- Payment gateway: Third-party LKR-compatible gateway (e.g. PayHere, Stripe LK)
- Delivery API: PickMe Partner API / Uber Delivery API
- Email/SMS: Third-party notification service (e.g. SendGrid, Twilio)

2.5 Constraints

- MVP must be delivered within 6 months of project initiation.
- Web platform only — no native mobile app in Phase 1.
- Payment processing must be in LKR only.
- PickMe/Uber delivery available only within their published coverage zones.
- Must comply with Sri Lanka consumer protection and personal data regulations.

2.6 Assumptions and Dependencies

- PickMe and/or Uber will sign an API partnership agreement by Month 2.
- A minimum of 10 pickup point partner locations will be contracted before launch.
- Third-party payment gateway supporting LKR will be procured by Month 1.
- At least 20 verified sellers will be onboarded before go-live.
- Customers have internet access via mobile data or broadband.

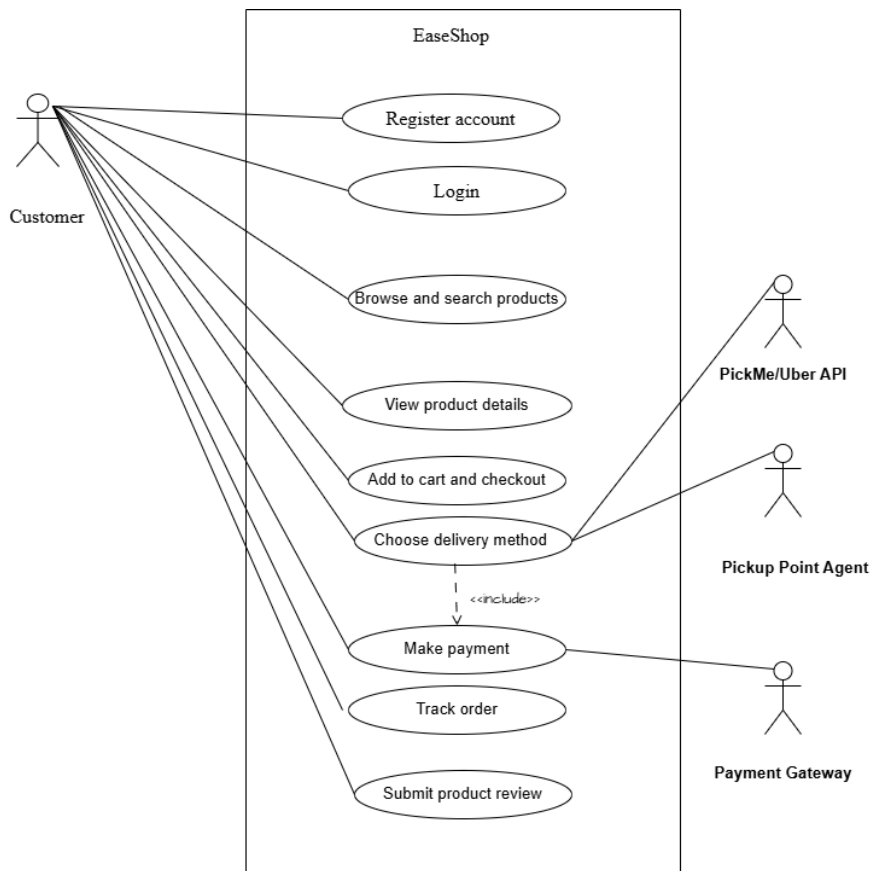
3. Use Case Diagrams

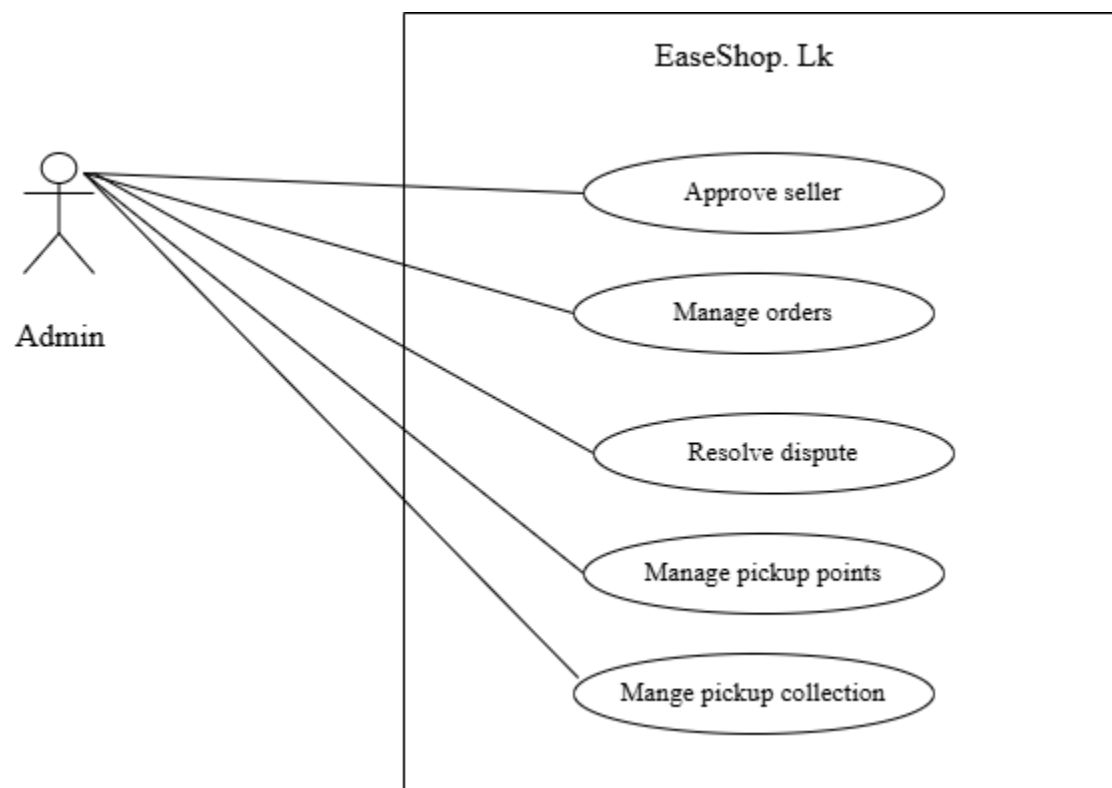
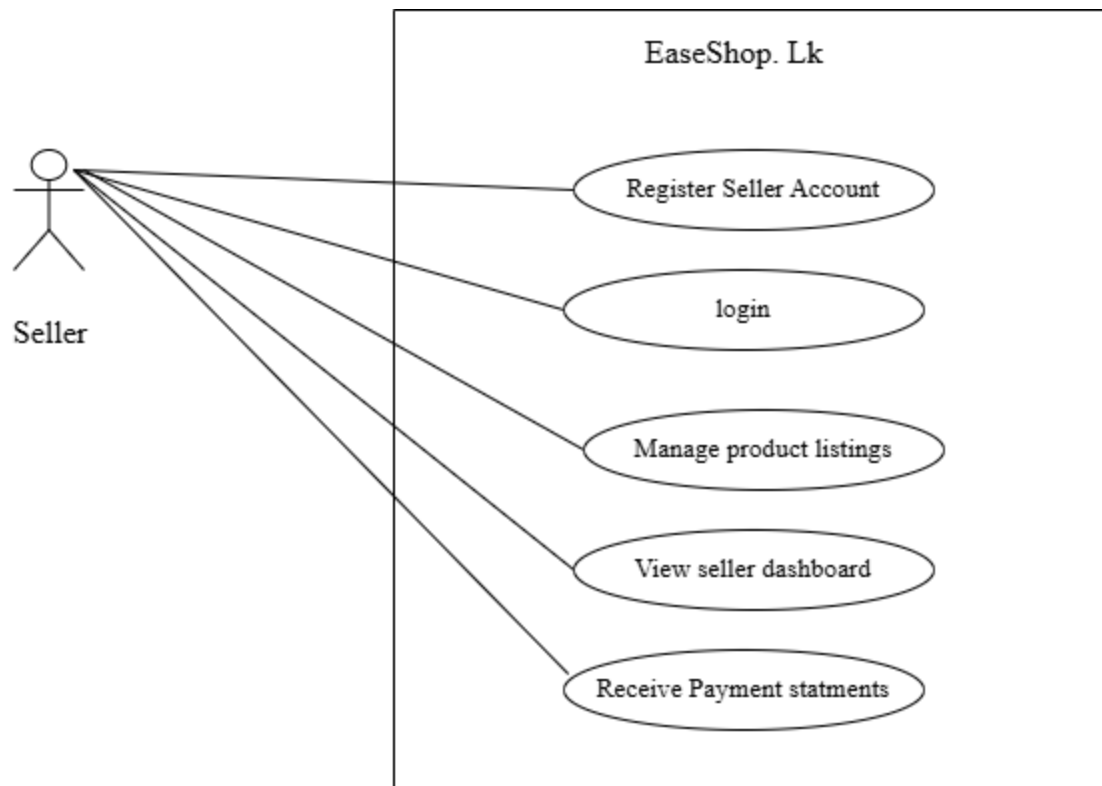
The following use case diagram identifies all actors and their interactions with the ShopEase LK system. As this is a text-based document, the diagram is represented in structured tabular form. A UML diagram image should be attached as Appendix A.

3.1 Actors

Actor	Description
Customer	Registers, browses products, places orders, tracks delivery, submits reviews
Seller	Registers, lists products, manages inventory, views sales analytics
Admin	Approves sellers, manages platform, resolves disputes, views analytics
PickMe/Uber API	System actor — provides real-time driver availability, ETA, price, and delivery status
Payment Gateway	System actor — processes card payments and confirms transactions
Pickup Point Agent	Confirms order collection at partner pickup location

3.2 Use Case Summary





UC ID	Use case name	Actor(s)	Description
UC-01	Register account	Customer	Create new customer account
UC-02	Login	Customer, Seller, Admin	Authenticate into the system
UC-03	Browse and search products	Customer	Find products by keyword, category, filter
UC-04	View product details	Customer	See product images, description, reviews, rating
UC-05	Add to cart and checkout	Customer	Place an order with delivery and payment selection
UC-06	Select delivery option	Customer, PickMe/Uber API	Choose urgent, standard, or pickup delivery
UC-07	Make payment	Customer, Payment Gateway	Pay by card or select COD
UC-08	Track order	Customer	View real-time delivery status and ETA
UC-09	Submit product review	Customer	Leave rating, text, photo, or video review
UC-10	Register as seller	Seller	Apply to become a verified seller
UC-11	Manage product listings	Seller	Add, edit, delete products and manage stock
UC-12	View seller dashboard	Seller	See orders, revenue, and inventory
UC-13	Approve seller	Admin	Verify and approve or reject seller application
UC-14	Manage orders	Admin	View and manage all platform orders
UC-15	Resolve dispute	Admin	Handle customer complaints and issue refunds

UC-16	Manage pickup points	Admin	Add, activate, or deactivate pickup locations
UC-17	Confirm pickup collection	Pickup Point Agent	Mark order as collected by customer

4. Main Use Case Scenarios

Each use case scenario follows the standard template: preconditions, main flow, alternate flow, and exception flow.

UC-05: Add to Cart and Checkout

Attribute	Detail
Use case ID	UC-05
Use case name	Add to cart and checkout
Actor	Customer
Preconditions	Customer is logged in. Product is in stock. Customer has a saved delivery address.
Post conditions	Order is created in the system. Payment is confirmed or COD is registered. Seller is notified.
Main flow	1. Customer clicks Add to Cart on a product page. 2. System adds item to cart and shows updated cart count. 3. Customer clicks Proceed to Checkout. 4. System displays saved addresses; customer selects or adds one. 5. System presents delivery options based on urgency selection and location. 6. Customer selects delivery option and confirms. 7. System shows order summary with total cost and delivery fee. 8. Customer selects payment method (card or COD). 9. If card: system redirects to payment gateway. 10. Gateway confirms payment. 11. System creates order and sends confirmation email and SMS.
Alternate flow	A1 (COD selected): Step 8 — customer selects COD. System skips payment gateway and registers COD against the order. Proceeds to step 11. A2 (New address): Step 4 — customer clicks Add New Address, enters details, system saves and selects it.
Exception flow	E1 (Payment failure): Payment gateway returns failure. System displays error and returns customer to payment selection. Order is not created. E2 (Item out of stock at checkout): System detects stock change, alerts customer, removes item from cart.

UC-06: Select Delivery Option

Attribute	Detail
Use case ID	UC-06
Use case name	Select delivery option
Actor	Customer, PickMe/Uber API (system actor)
Preconditions	Customer is at checkout. Delivery address is confirmed.
Postconditions	Delivery method is recorded against the order. Delivery cost is added to the order total.
Main flow	1. System asks: Is this delivery urgent? 2. Customer selects Urgent or Standard. 3a. If Urgent AND within PickMe/Uber coverage: system calls PickMe/Uber API; API returns available drivers, ETA, and price; system displays PickMe and Uber as options with live ETA and fee. 3b. If Urgent AND outside coverage: system shows ShopEase delivery and pickup points only with message. 3c. If Standard AND order > Rs 3,000: system shows ShopEase free delivery as recommended option. 3d. If Standard AND order < Rs 3,000: system shows all options with fees. 4. Pickup point option is always shown. 5. Customer selects preferred option. 6. System confirms selection and adds delivery fee to total.
Alternate flow	A1: Customer changes selection — system recalculates delivery fee and updates total.
Exception flow	E1: PickMe/Uber API timeout — system shows ShopEase delivery and pickup points with a message that courier options are temporarily unavailable.

UC-10: Register as Seller

Attribute	Detail
Use case ID	UC-10
Use case name	Register as seller
Actor	Seller, Admin
Preconditions	Applicant has a valid business or NIC number and bank account.

Postconditions	Seller account is pending admin verification. Seller receives confirmation email.
Main flow	1. Seller clicks Register as Seller. 2. System displays seller registration form. 3. Seller enters: name, business name, NIC/business registration, bank account details, contact number, product category. 4. Seller submits form. 5. System validates all fields. 6. System creates seller account with status Pending. 7. System notifies Admin of new application. 8. Admin reviews and approves or rejects. 9. System sends approval/rejection email to seller. 10. If approved: seller account is activated and verified badge is assigned.
Exception flow	E1: Duplicate NIC or business number detected — system shows error and rejects submission. E2: Admin rejects application — seller receives rejection email with reason.

UC-15: Resolve Dispute

Attribute	Detail
Use case ID	UC-15
Use case name	Resolve dispute
Actor	Admin, Customer, Seller
Preconditions	Customer has submitted a dispute request against an order.
Main flow	1. Customer submits dispute via order history with reason and evidence (photos). 2. System notifies Admin and Seller. 3. Admin reviews dispute details, customer evidence, and seller response. 4. Admin decides: approve refund, partial refund, or reject. 5. If refund approved: system processes refund to original payment method or issues store credit for COD orders. 6. System notifies customer and seller of decision.
Exception flow	E1: Seller does not respond within 3 business days — system auto-escalates to Admin for decision.

5. Functional requirements

All requirements use 'shall' (mandatory), 'should' (recommended), or 'may' (optional) per IEEE 830 convention. Each requirement is uniquely identified and linked to a use case and source stakeholder.

5.1 User Registration and Authentication

ID	Requirement (system shall...)	Priority	UC ref	Source
FR-01	Allow customers to register using email address or mobile number.	Must	UC-01	CEO, Customer
FR-02	Send a verification email or SMS OTP upon registration.	Must	UC-01	IT Manager
FR-03	Allow customers to log in using email/mobile and password.	Must	UC-02	Customer
FR-04	Provide a password reset function via registered email or mobile.	Must	UC-02	Customer
FR-05	Allow customers to save multiple delivery addresses in their profile.	Must	UC-01	Customer
FR-06	Support separate login portals for customers, sellers, and admins.	Must	UC-02	IT Manager

5.2 Product Catalog and Search

ID	Requirement (system shall...)	Priority	UC ref	Source
FR-07	Display products grouped by category on the home page.	Must	UC-03	Customer
FR-08	Allow customers to search products by keyword, category, price range, and seller rating.	Must	UC-03	Customer
FR-09	Display product pages with minimum 3 photos, description, price, stock status, and seller badge.	Must	UC-04	Customer, Seller
FR-10	Show average product rating (1-5 stars) on product listing and detail pages.	Must	UC-04	Customer
FR-11	Display seller verified badge on all product listings from approved sellers.	Must	UC-04	CEO, Customer
FR-12	Allow customers to add products to a wishlist.	Should	UC-03	Customer

5.3 Shopping Cart and Checkout

ID	Requirement (system shall...)	Priority	UC ref	Source
FR-13	Allow customers to add, remove, and update quantities in the shopping cart.	Must	UC-05	Customer
FR-14	Display running cart total including delivery fee at all checkout stages.	Must	UC-05	Customer
FR-15	Allow application of promo/discount codes at checkout with validated discount applied.	Should	UC-05	Marketing
FR-16	Prevent checkout if any item in the cart is out of stock; notify customer.	Must	UC-05	IT Manager

5.4 Delivery Selection

ID	Requirement (system shall...)	Priority	UC ref	Source
FR-17	Present a delivery urgency selection (Urgent / Standard) during checkout.	Must	UC-06	CEO
FR-18	If Urgent selected and customer is within PickMe/Uber coverage, call PickMe/Uber API and display options with real-time ETA and price.	Must	UC-06	CEO, IT Manager
FR-19	If Urgent selected and customer is outside coverage, display ShopEase delivery and pickup points only, with explanatory message.	Must	UC-06	IT Manager
FR-20	If Standard selected and order value exceeds Rs 3,000, offer ShopEase delivery as free option.	Must	UC-06	CEO
FR-21	Always display pickup point option to all customers regardless of urgency or order value.	Must	UC-06	CEO, Customer
FR-22	Display estimated delivery time and fee for each delivery option before customer confirms.	Must	UC-06	Customer
FR-23	If PickMe/Uber API times out, display ShopEase and pickup options with a temporary unavailability notice.	Must	UC-06	IT Manager

5.5 Payment

ID	Requirement (system shall...)	Priority	UC ref	Source
FR-24	Process card payments (Visa, Mastercard) in LKR via integrated payment gateway.	Must	UC-07	CEO, IT Manager
FR-25	Offer cash-on-delivery as a payment option for all three delivery methods.	Must	UC-07	Customer
FR-26	For COD orders, generate a unique collection code to be confirmed by driver or pickup agent.	Must	UC-07	CS Lead, IT Manager
FR-27	Send payment confirmation email and SMS immediately after successful card transaction.	Must	UC-07	Customer
FR-28	Prevent duplicate order submission on payment gateway redirect.	Must	UC-07	IT Manager
FR-29	Transfer seller payment within 7 business days of delivery confirmation.	Must	UC-07	Seller

5.6 Order Management and Tracking

ID	Requirement (system shall...)	Priority	UC ref	Source
FR-30	Generate a unique order ID upon order confirmation.	Must	UC-08	IT Manager
FR-31	Display real-time order status (Confirmed, Packed, Dispatched, Out for Delivery, Delivered).	Must	UC-08	Customer
FR-32	Send automated SMS and email notification at each order status change.	Must	UC-08	Customer, CS Lead
FR-33	Show live driver location on map for PickMe/Uber deliveries.	Should	UC-08	Customer
FR-34	Allow customers to view full order history in their profile.	Must	UC-08	Customer

5.7 Reviews and Ratings

ID	Requirement (system shall...)	Priority	UC ref	Source
FR-35	Allow customers to submit a product review only after confirmed delivery.	Must	UC-09	Customer

FR-36	Support text, photo, and video reviews up to 30 seconds.	Should	UC-09	Customer, Marketing
FR-37	Calculate and update the product average rating after each new review.	Must	UC-09	Customer
FR-38	Allow Admin to remove reviews that violate platform policy.	Must	UC-09	Admin

5.8 Seller Management

ID	Requirement (system shall...)	Priority	UC ref	Source
FR-39	Provide a seller registration form capturing: name, business name, NIC/business reg, bank details, product category.	Must	UC-10	CEO, Seller
FR-40	Set newly registered sellers to Pending status until admin approval.	Must	UC-10	CEO
FR-41	Assign a verified badge to approved sellers, displayed on all their listings.	Must	UC-10, UC-11	CEO, Customer
FR-42	Allow sellers to add, edit, and delete product listings with photos, price, description, and stock count.	Must	UC-11	Seller
FR-43	Show sellers a dashboard with: total sales, pending orders, stock alerts, and revenue chart.	Must	UC-12	Seller
FR-44	Alert seller when product stock falls below 5 units.	Should	UC-12	Seller

5.9 Admin Panel

ID	Requirement (system shall...)	Priority	UC ref	Source
FR-45	Allow Admin to review, approve, or reject seller applications with a reason.	Must	UC-13	CEO
FR-46	Allow Admin to view and manage all platform orders.	Must	UC-14	CEO
FR-47	Allow Admin to resolve customer disputes and issue full or partial refunds.	Must	UC-15	CS Lead
FR-48	Allow Admin to add, activate, and deactivate pickup point locations.	Must	UC-16	CEO

FR-49	Provide Admin with a platform analytics dashboard: total revenue, active users, top sellers, order volumes.	Must	UC-14	CEO
FR-50	Allow Admin to deactivate seller accounts for policy violations.	Must	UC-13	CEO

6. Non Fictional Requirements

6.1 Performance

NFR ID	Requirement	Target metric
NFR-01	Product listing pages shall load within acceptable time under normal user load	< 3 seconds
NFR-02	Search results shall be returned after query submission	< 2 seconds
NFR-03	Checkout process shall complete from cart to confirmation	< 5 seconds
NFR-04	PickMe/Uber API call shall return results during checkout	< 3 seconds
NFR-05	Platform shall support concurrent users at launch without degradation	> 1,000 users

6.2 Security

NFR ID	Requirement	Standard/ measure
NFR-06	All data transmission between browser and server shall be encrypted	HTTPS / TLS 1.2+
NFR-07	Customer passwords shall be stored using a one-way hashing algorithm	bcrypt
NFR-08	Payment card data shall not be stored on ShopEase servers	PCI-DSS via gateway
NFR-09	All API keys (PickMe/Uber, payment gateway) shall be stored in environment variables	No hardcoding
NFR-10	System shall lock accounts after 5 consecutive failed login attempts	30-minute lockout

6.3 Availability and Reliability

NFR ID	Requirement	Target
NFR-11	Platform shall maintain uptime during core shopping hours	99% uptime
NFR-12	Scheduled maintenance windows shall not occur during peak hours (6pm-10pm)	Off-peak only

NFR-13	System shall automatically back up all order and customer data	Daily backups
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6.4 Usability

NFR ID	Requirement	Target
NFR-14	A new customer shall be able to complete first purchase without training	< 5 min task time
NFR-15	Platform shall display clear error messages in plain language for all validation failures	All forms
NFR-16	Platform shall be responsive and usable on mobile browsers without a native app	320px+ viewport

6.5 Scalability and Maintainability

NFR ID	Requirement	Target
NFR-17	System architecture shall support horizontal scaling to accommodate user growth	Cloud auto-scale
NFR-18	Codebase shall follow modular architecture to allow Phase 2 features to be added without full rewrite	Modular design

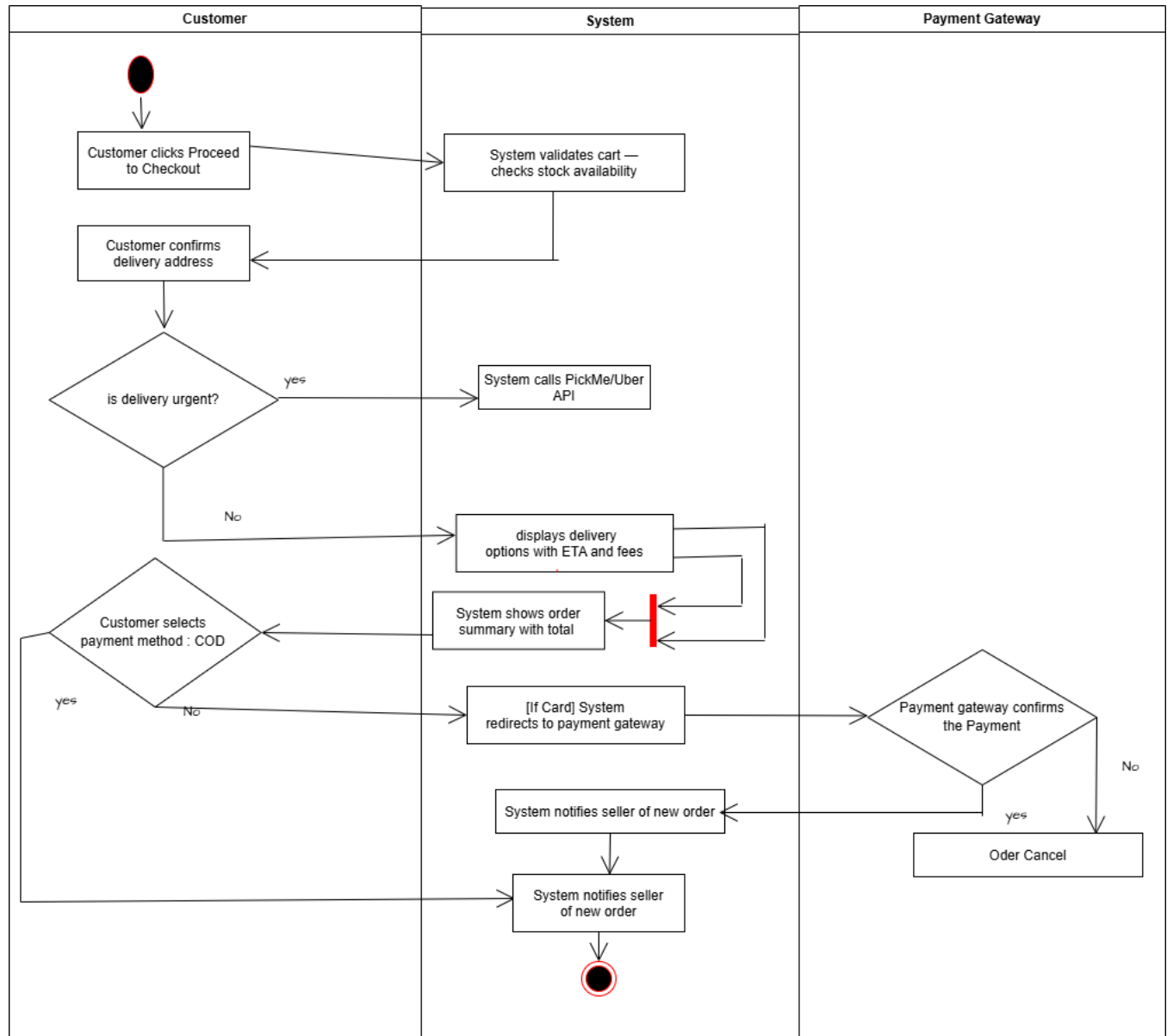
6.6 Compliance

NFR ID	Requirement	Regulation
NFR-19	Platform shall comply with Sri Lanka consumer protection laws for returns and refunds	SL Consumer Law
NFR-20	Platform shall not share customer personal data with third parties without consent	Data Privacy

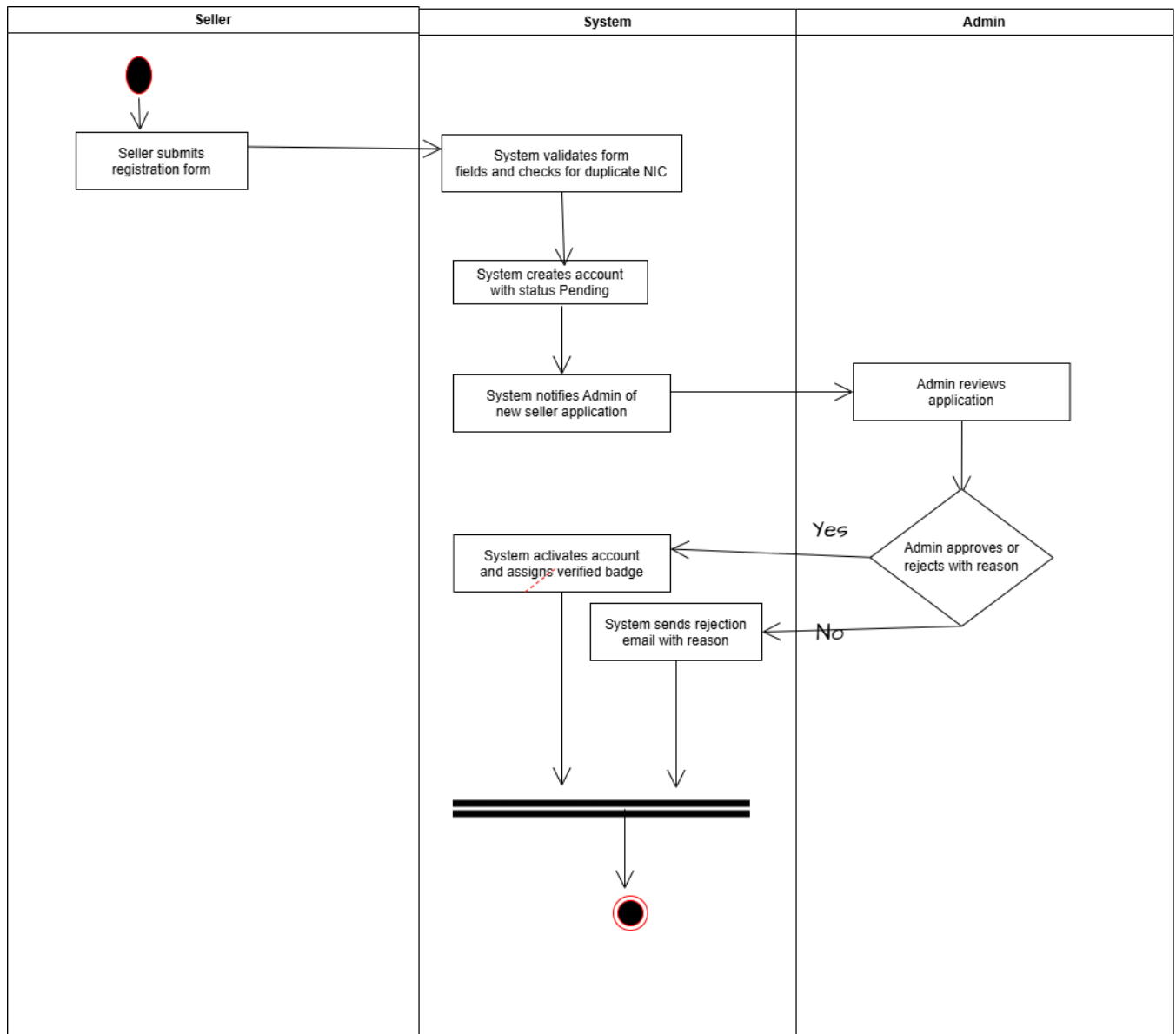
7. Activity Diagram

Activity diagrams describe the step-by-step flow of key system processes. A UML diagram image for each should be attached in Appendix B. The flows below are the textual representation for this document.

7.1 Customer Checkout Activity Flow



7.2 Seller Onboarding Activity Flow



8. Data Design (Database Design)

The following entity definitions describe the core data structure for the ShopEase LK platform. An Entity Relationship Diagram (ERD) image should be attached as Appendix C. Relationships between entities are described in Section 8.2.

Note: Including the data design in this SRS is intentional for this MVP project to provide developers with a complete single-reference document. In a larger organization this would typically reside in a separate System Design Document (SDD).

8.1 Entity Definitions

CUSTOMER

Field name	Data type	Key	Description
customer_id	INT	PK	Unique customer identifier
full_name	VARCHAR(100)		Customer full name
email	VARCHAR(100)	UNIQUE	Login email address
mobile	VARCHAR(15)	UNIQUE	Mobile number for SMS
password_hash	VARCHAR(255)		bcrypt hashed password
is_verified	BOOLEAN		Email/mobile verification status
created_at	DATETIME		Account creation timestamp

ADDRESS

Field name	Data type	Key	Description
address_id	INT	PK	Unique address identifier
customer_id	INT	FK(CUSTOMER)	Owner of address

label	VARCHAR(50)		Home, Office, etc.
street	VARCHAR(200)		Street and number
city	VARCHAR(100)		City
district	VARCHAR(100)		District
is_default	BOOLEAN		Default delivery address flag

SELLER

Field name	Data type	Key	Description
seller_id	INT	PK	Unique seller identifier
business_name	VARCHAR(150)		Trading name
nic_or_brn	VARCHAR(50)	UNIQUE	NIC or business registration
bank_account	VARCHAR(50)		Settlement bank account
contact_mobile	VARCHAR(15)		Contact number
status	ENUM		Pending, Approved, Rejected, Suspended
is_verified	BOOLEAN		Verified badge flag
created_at	DATETIME		Registration timestamp

PRODUCT

Field name	Data type	Key	Description
product_id	INT	PK	Unique product identifier
seller_id	INT	FK(SELLER)	Seller who owns listing

category_id	INT	FK(CATEGORY)	Product category
name	VARCHAR(200)		Product name
description	TEXT		Full product description
price	DECIMAL(10,2)		Unit price in LKR
stock_count	INT		Available units
avg_rating	DECIMAL(3,2)		Average review rating
is_active	BOOLEAN		Listing visibility flag
created_at	DATETIME		Listing creation date

ORDER

Field name	Data type	Key	Description
order_id	INT	PK	Unique order identifier
customer_id	INT	FK(CUSTOMER)	Ordering customer
address_id	INT	FK(ADDRESS)	Delivery address
delivery_type	ENUM		Urgent, Standard, Pickup
delivery_partner	ENUM		PickMe, Uber, ShopEase, Pickup
pickup_point_id	INT	FK(PICKUP_POINT)	Pickup location if applicable
payment_method	ENUM		Card, COD
payment_status	ENUM		Pending, Confirmed,

			Failed, Refunded
order_status	ENUM		Confirmed, Packed, Dispatched, Delivered, Disputed
total_amount	DECIMAL(10,2)		Total incl. delivery fee
delivery_fee	DECIMAL(10,2)		Delivery cost
cod_code	VARCHAR(10)		Unique COD collection code
created_at	DATETIME		Order placement timestamp

ORDER_ITEM

Field name	Data type	Key	Description
item_id	INT	PK	Unique order item identifier
order_id	INT	FK(ORDER)	Parent order
product_id	INT	FK(PRODUCT)	Ordered product
quantity	INT		Units ordered
unit_price	DECIMAL(10,2)		Price at time of order
seller_id	INT	FK(SELLER)	Seller fulfilling this item

REVIEW

Field name	Data type	Key	Description
review_id	INT	PK	Unique review identifier
product_id	INT	FK(PRODUCT)	Reviewed product
customer_id	INT	FK(CUSTOMER)	Reviewer

order_id	INT	FK(ORDER)	Verified purchase order
rating	TINYINT		1 to 5 star rating
review_text	TEXT		Written review content
media_urls	TEXT		Photo/video URLs (JSON array)
created_at	DATETIME		Review submission timestamp

DELIVERY

Field name	Data type	Key	Description
delivery_id	INT	PK	Unique delivery record
order_id	INT	FK(ORDER)	Associated order
partner_ref	VARCHAR(100)		PickMe/Uber job reference ID
status	ENUM		Assigned, Picked Up, In Transit, Delivered
driver_name	VARCHAR(100)		Assigned driver name
driver_mobile	VARCHAR(15)		Driver contact (PickMe/Uber)
estimated_eta	DATETIME		ETA from delivery partner API
delivered_at	DATETIME		Actual delivery timestamp

PICKUP_POINT

Field name	Data type	Key	Description
pickup_point_id	INT	PK	Unique pickup location ID
name	VARCHAR(150)		Shop/location name
address	VARCHAR(250)		Full address
district	VARCHAR(100)		District for filtering
contact_mobile	VARCHAR(15)		Agent contact number
is_active	BOOLEAN		Location operational status

CATEGORY

Field name	Data type	Key	Description
category_id	INT	PK	Unique category identifier
name	VARCHAR(100)		Category name
parent_category_id	INT	FK(CATEGORY)	Parent for sub-categories (nullable)

8.2 Entity Relationships

Entity A	Relationship	Entity B
CUSTOMER	1 to many	ADDRESS — one customer can have many addresses
CUSTOMER	1 to many	ORDER — one customer can place many orders
SELLER	1 to many	PRODUCT — one seller can have many product listings

ORDER	1 to many	ORDER_ITEM — one order can have many items
PRODUCT	1 to many	ORDER_ITEM — one product can appear in many orders
ORDER	1 to 1	DELIVERY — one order has one delivery record
PRODUCT	1 to many	REVIEW — one product can have many reviews
CUSTOMER	1 to many	REVIEW — one customer can write many reviews
ORDER	Many to 1	PICKUP_POINT — many orders can use the same pickup point
CATEGORY	1 to many	PRODUCT — one category contains many products

9. External Interface Requirements

9.1 User Interface Requirements

- The platform shall use a responsive web design that adapts to screen widths from 320px (mobile) to 1440px (desktop).
- Navigation shall include: Home, Categories, Cart, My Orders, and My Account at all times.
- All forms shall display inline validation messages immediately on blur.
- Error messages shall be written in plain English without technical codes.
- The checkout flow shall be a maximum of 4 steps: Cart, Address, Delivery, Payment.
- Product images shall support zoom on hover (desktop) and pinch-zoom (mobile).

9.2 PickMe/Uber API Interface

Attribute	Detail
API type	REST API — JSON request/response
Authentication	OAuth 2.0 Bearer token
Request (delivery quote)	POST customer location, destination address, package weight/size
Response	Available driver count, ETA (minutes), price quote (LKR), coverage zone status
Webhook	ShopEase receives delivery status updates: Assigned, Picked Up, In Transit, Delivered
Timeout handling	If no response within 3 seconds, fallback to ShopEase delivery and pickup options
Coverage check	System calls coverage endpoint with customer lat/lng before presenting courier options

9.3 Payment Gateway Interface

Attribute	Detail
Gateway type	Third-party LKR-compatible gateway (e.g. PayHere)
Integration method	Redirect-based — customer redirected to gateway hosted page
Data passed	Order ID, amount (LKR), customer name, email, return URL
Response	Payment status: SUCCESS, FAILED, CANCELLED — returned via webhook and redirect

Security	ShopEase never handles raw card data — all processed by gateway (PCI-DSS compliant)
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9.4 Notification Service Interface

Attribute	Detail
Providers	Email: SendGrid or SMTP. SMS: Twilio or Dialog Axiata API
Triggers	Registration, order confirmation, payment confirmation, each status change, delivery completion
Fallback	If SMS fails, system retries once; if email fails, system logs and alerts Admin

10. System Constraints

Constraint type	Description
Technology	Web platform only — no native mobile app in Phase 1. Must support Chrome, Firefox, Safari, Edge.
Payment	All transactions in LKR only. Card data must not be stored on ShopEase infrastructure.
Delivery API	PickMe/Uber delivery is limited to their published coverage zones. Coverage cannot be extended unilaterally.
Timeline	MVP must be fully developed, tested, and launched within 6 months of project start.
Legal	Platform must comply with Sri Lanka Consumer Protection Act and any applicable personal data regulations.
Language	Phase 1 English only. Sinhala and Tamil localisation deferred to Phase 2.
Infrastructure	Hosting must be cloud-based to support auto-scaling. On-premise hosting is not permitted.

Appendix A — Sign-off

Name	Role	Date	Signature
	Business Analyst		
	Product Owner / Sponsor		
	IT Manager		

